# Equation Sheet

$$work = force \times distance$$

$$acceleration = \frac{force}{mass} \qquad speed = \frac{distance}{time}$$

$$kinE = \frac{1}{2} \times mass \times (speed)^2 \qquad gravE = weight \times height$$

$$input = output + system\ energy\ changes$$

$$energy = power\ x\ time \qquad power = \frac{energy}{time}$$

$$input = work + exhaust \qquad work = input - exhaust$$

$$efficiency = \frac{useful\ output}{energy\ input}$$